

Manual

Failure Mode Analysis / Measure Tracking FEP-FSM 8.1

Version 1.1.1374

Last changed on: 19.06.2020

Copyright

©Copyright 2020 All rights reserved.

SAP® and R/3® are registered trademarks of SAP AG.

WINDOWS® is a registered trademark of Microsoft Corporation.

MPDV® and HYDRA® are registered trademarks of MPDV Mikrolab GmbH.

ORACLE® is a registered trademark of ORACLE Corporation, California, USA.

Copying and distribution of this documentation or any part thereof, for any purpose or in any form, is prohibited without prior written permission from MPDV Mikrolab GmbH.

The information contained in this documentation is subject to change without prior notice.

Contents

1	Failure Mode Analysis / Measure Tracking ..	Error! Bookmark not defined.
2	Failure Mode Analysis.....	5
3	Measure Tracking	12

1 Failure Mode Analysis / Measure Tracking

Purpose

The component is employed when failure modes are to be identified on the basis of the recorded faults in order to be able to initiate selective improvements in order to raise the product quality.

The tracking of measures allows central processing of measures and selective filtering, e.g. by incomplete measures.

Implementation Notes

Use of this component is to be recommended where as part of the inspection data collection, errors are recorded or the automatic generation of errors is used and measures are initiated via the AIP.

Integration

The component "Inspection planning for in-process inspections" with the corresponding inspection requirements and the errors and measures recorded for the inspection results form the basis, whereby the inspection data must first be collected using the AIP collection client.

Features

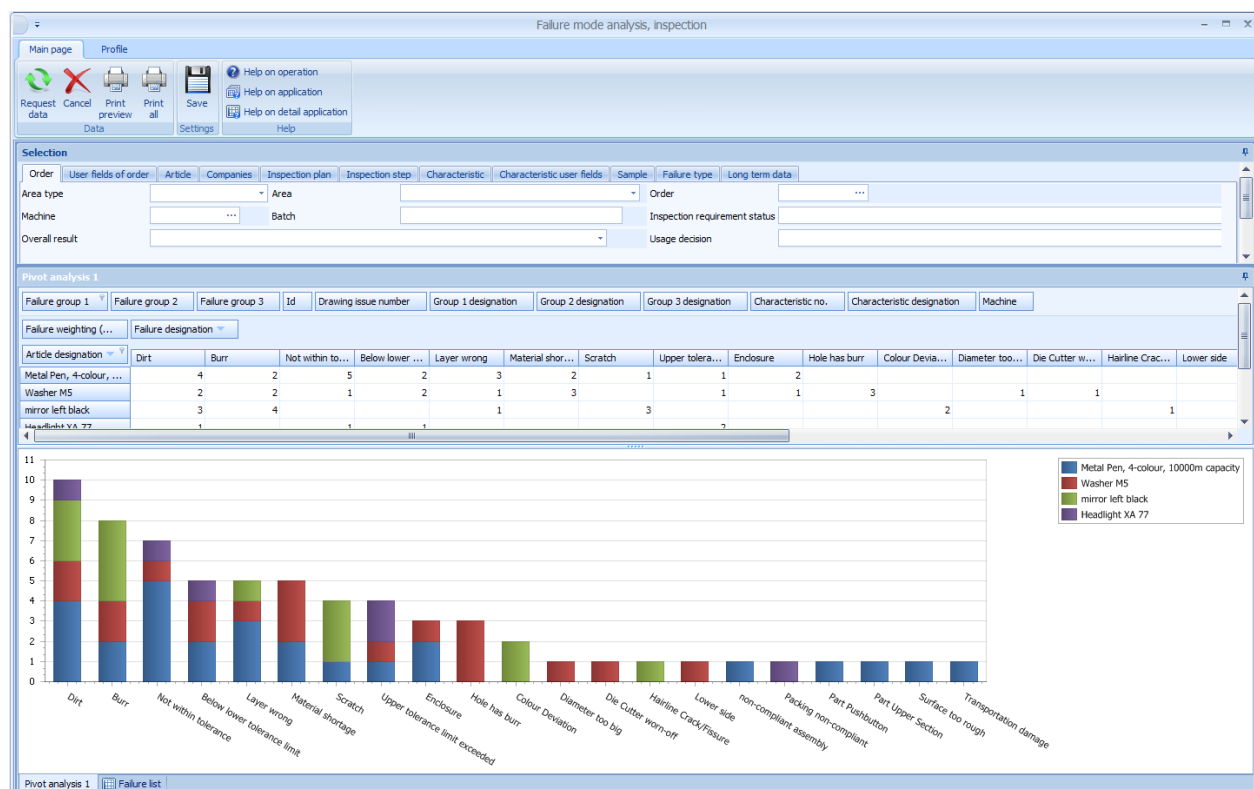
The component offers the following functions:

- Graphic evaluation of the recorded error types, locations and causes with extensive filter criteria and representation possibilities in the Form of a pivot analysis
- Tracking of measures across all orders with editing possibilities (status, effectiveness, degree of fulfillment, actual completion date, comment on the activities performed, filtering by responsible persons, etc.)

2 Failure Mode Analysis

Summary

Menu	Quality management → QM evaluation → Failure mode analysis
Transaction code	faep
Function authorization	faep



Utilization

The failure mode analysis allows for failures of the following types recorded during the inspection process or generated automatically to be evaluated.

- Failure type (FT)
- Failure location (FL) and
- Failure cause (FC).

In this context evaluations/reports are based on pivot functions. They allow for the distribution of failure types (frequency) to be presented for each article/item referring to a period of time that has been filtered before. These analyses help determine the core areas that might require action to be taken to optimize the production process with respect to quality.

Integration

The failure mode analysis evaluates failures of inspection requirements from the area types:

- goods receipt,
- initial sample inspection,
- production and
- goods issue

Consequently, the failure mode analysis is a comprehensive analysis tool.

Prerequisite

There are no special requirements. The only prerequisite is that failures have to be recorded, which, in turn, need to be defined within the master data of quality management before.

Selection criteria

The paragraph that follows shows some of the available selection criteria. Self-explanatory filter options are not listed.

"Order" tab

Area type

Selection list of area types

- in-production inspection
- goods receipt inspection
- goods issue inspection
- initial sample inspection

The list of area types may be restricted subject to the respective licenses in use.

Area

Selection list of the configured areas of the previously filtered area type. By default, the following areas are available. .

- In-production inspection: production
- Goods receipt inspection: goods receipt
- Goods issue inspection: goods issue
- Initial sample inspection: initial sample

Further areas can be generated through customizing.

Inspection request status

Selection list including multiple selection options for all inspection request statuses

Overall result

Selection list for the results of inspection requests (pass, fail, conditionally pass)

Usage decision

Selection list of usage decisions (e.g. release, conditional release, rework, reject, sort) that are available, once an inspection request has been completed.

"Inspection point identification" tab

If these fields are not available, you require a new program version of this application.

Workplace

Workplace number

Partial batch

Partial batch number

Batch

Batch number

Field 1...3

Additional fields for inspection points

"Additional fields for inspection points" tab

If these fields are not available, you require a new program version of this application.

Field 4...8

Additional fields for inspection points



The fields "field 1" to "field 8" are enabled subject to MPDV customizing and are assigned individual names. As these field names are flexible, they are only entitled "field 1" to "field 8" in this document.

Field 1 includes, for example, the tool if cavity-related data collection is enabled.

"Sample" tab

Machine sample

Selection list of machines/workplaces

The machine assigned to the first measured value is filtered here, provided that different machines are assigned to the measured values of the same sample.

Toolbar

There are no other special function buttons in addition to the standard functions/features.

"Graphic failure analysis" detail applications

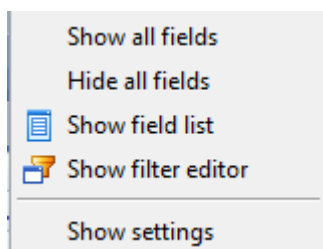
Data is displayed in a pivot table in combination with bar charts. Different application functions are provided for the presentation. The failures that have been filtered beforehand are the data basis.

The general pivot functions are not described in more detail in this document. The paragraphs that follow only describe the elementary functions of this evaluation/report.

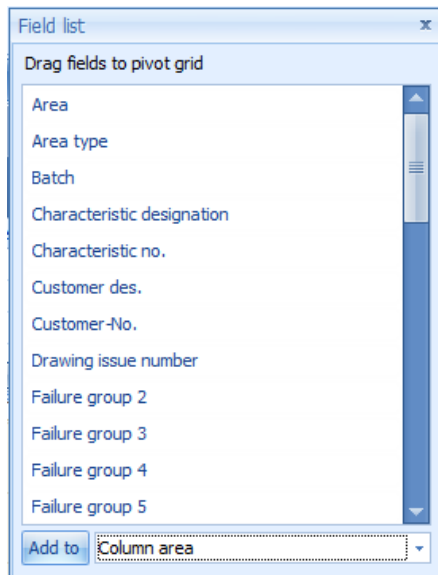
Pivot evaluations/reports provide the following benefits.

- Large amounts of data may quickly be summarized and presented.
- Rows and columns can be exchanged to have the source data summarized differently.
- Simple filters by "drag and drop" with additional detail filters.
- Due to an interactive way of representation, data can be summarized and analyzed in different formats and using different calculation methods.

The below context menu can be opened by clicking the right mouse button.

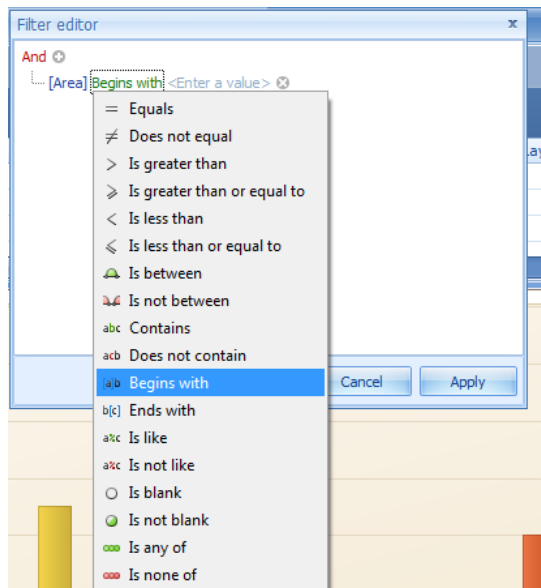


The function "show field list" allows for the fields that are to be used in the pivot analysis to be selected. The below figure shows a possible field list.

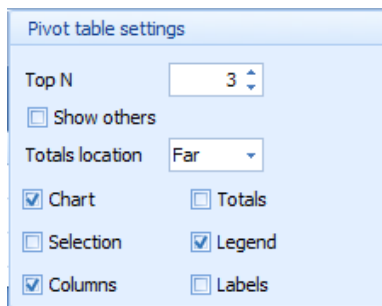


The requested fields may be put into the evaluation area by drag & drop.

In addition to the selection criteria, the "show filter editor" function enables further flexible restrictions of the data basis.

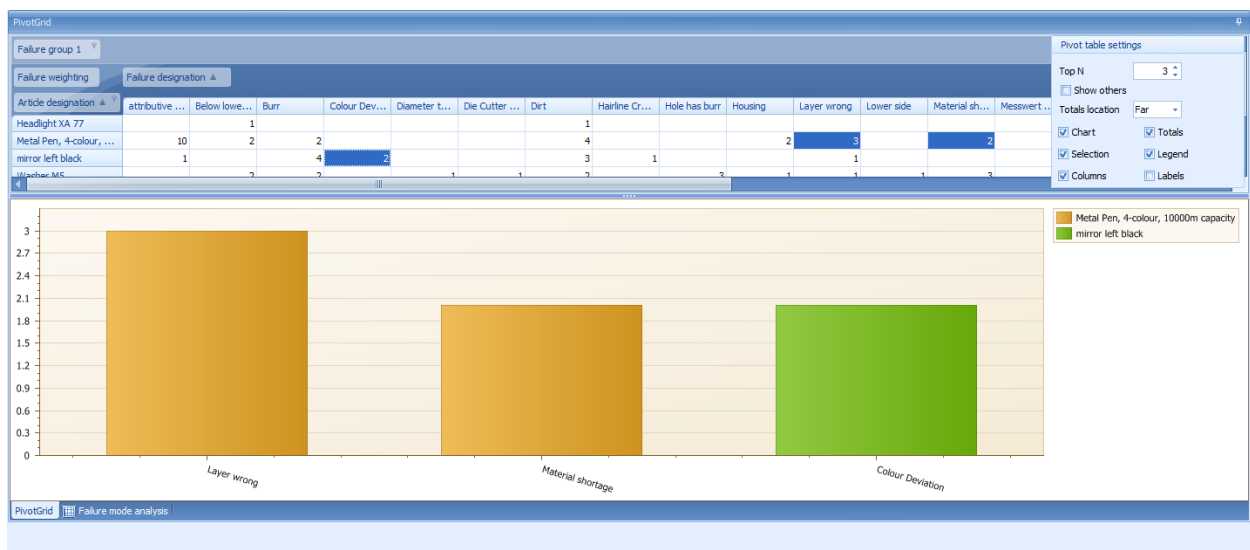


The below dialog is opened to show the settings made.



If the "selection" option is checked entire areas may be selected in the table view. In this case, the graphic representation is based on the selected rows. If the "label" option is checked it is possible to show the total number of each bar.

The below figure explains these functions.



The row showing the total result may be displayed additionally in the bar chart if the "totals" option is checked. Provided that the "selection" function is checked and the corresponding cells of the "total result" row are selected, the total result of the corresponding column is added to the respective bar.

It is switched between the graphic presentation of the corresponding number of columns or rows by checking/unchecking the "columns" option.

"Failure list" detail applications

The failure list shows the failures including the referenced data that are filtered based on the used selection criteria. Normally, the referenced data corresponds to the field list for the pivot analysis.

3 Measure Tracking

Summary

Menu	Quality management --> QM evaluation --> Measure tracking Quality management → Complaint management → Measure tracking
Transaction code	tm
Function authorization	tm

The measure tracking function provides an overview of all generated measures. The "type" specifies from which area/application the measure derives. The application allows for measures to be edited at a central place and assigned statuses to be changed.

Utilization

Filtering by the "status" is the central element allowing to select, for example, all measures that are open, done, read or in process. If this is combined with filtering or grouping by the party responsible, it is possible to create a list showing all measures that are still open by person, department, etc.

Measures may be changed. In addition to changing statuses, it is also allowed to enter a comment about the activities made in the fields "measure text" and/or "comment". Furthermore, this application also includes the actual date indicating when a measure has been completed and the degree of effectiveness and fulfillment in %. It is even allowed to change the party responsible.

The "reference" group provides information on the origin of the measure. If it is, for example, a measure generated in complaint management, the type, area and the complaint or complaint detail will be shown here.

The relevant application can be opened by clicking the corresponding button in the toolbar. The application is determined by the content of the "context" field.

The "long-term data" tab is only available if the archiving license has been purchased. If this option is set, long-term data will also be used when measures are displayed and tracked.

Integration

The measure tracking function is connected with all applications where measures are documented.

Prerequisite

There are no special requirements to be met.

Selection criteria

Except for the "reference" tab, selection criteria are self-explanatory and not described separately. The filter fields "key 1" up to "key 5" of the "reference" tab allow for measures to be restricted specifically to the origin of the measure. In case of a complaint measure, the field "key 1" includes, for example, the complaint number generated automatically in HYDRA. The key field 2 then includes the number of the complaint detail.

Field descriptions

The display dialog shows different tabs. However, if newly created/edited, there is only one tab (the "measure" tab) that includes all pieces of information.

Measures tab

Measure

Measure number and/or selection or direct entry of a measure number. The relevant master data catalog can be opened for selection purposes.

Measure designation

The designation of the assigned measure number is shown. If the measure number is input directly, the designation will only be shown upon saving.

Text

Free text field to enter a complementary measure text.

Comment

Free text field to enter a complementary comment for the measure.

Status

Available measure types can be displayed or selected. The types "in process", "read", "done" and "open" are provided by default.

Measure type

Available measure types can be displayed or selected. The following types are available by default "short-term", "medium-term", "long-term" and "no assignment".

Fulfillment [%]

Fulfillment in % can be displayed or entered.

Effectiveness [%]

Effectiveness in % can be displayed or entered.

External

This field can be used to control the printout of forms. However, it does not have a special function.

Context

Assigned object, e.g. inspection requirement, inspection step, complaint management, complaint detail, energy management

Type

The referenced application type can be displayed or selected, e.g. in-production inspection, goods receipt/goods issue inspection, initial sample inspection, test equipment inspection, complaint management, energy management

Area

The area of the type can be displayed or selected, e.g. production for in-production inspection

Key 1 to 5 (general)

The key fields are assigned depending on the context. These fields are empty for measures pertaining to energy management.

Key 1

Shows the internal and unique complaint number for measures from complaint management. The inspection requirement number is entered here for measures of inspection requirements or inspection steps.

Key 2

Shows the complaint detail number for measures from complaint management including assignment to the complaint detail. The inspection step number is entered here for measures of inspection requirements or inspection steps.

Key 3

Shows the internal, unique ID number of all failure analysis entries of a complaint for measures from complaint management that are also collected in failure analysis. The OP sequence number of the characteristic is entered here for measures of inspection requirements or inspection steps, provided that data in relation to the characteristic has been collected.

Key 4

This field is empty for measures from complaint management. The sample number is entered here for measures of inspection requirements or inspection steps, provided that data in relation to the sample number has been collected.

Key 5

This field is empty for measures from complaint management. The measured value number is entered here for measures of inspection requirements or inspection steps, provided that data in relation to the sample number has been collected.

Party in charge tab**Party in charge type**

Different types may be selected when inputting data. The type "external person" is used, as normally people are responsible for dealing with measures.

Party in charge

The party in charge is shown or it may be chosen from the list of responsible parties. Which entry is transferred to the list of responsible parties is defined within master data. The selected entry is used as the responsible party.

Party in charge name 1, name 2, name 3

The contents of the fields name 1, name 2 and name 3 of the responsible party are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

Dates tab**Target date/time**

A date and optionally a time by which the measure has to be finished may be displayed or entered. It will not be monitored automatically, whether or not this time limit is kept. The content of this field is fundamental to the "measures tracking" application, as it may be determined manually for all measures (global) which of them have exceeded the target date, for example.

Actual date/time

A date and optionally the time by which the measure has been finished may be displayed or entered. However, this field does not have a special function. In the "measure tracking" application, this field can be used to manually determine measures that have been completed with delay.

"Reference" tab

Area

Shows the referenced area within the application type, e.g. production

Toolbar

Apart from standard functions, the referenced measure, referenced inspection requirement or the consumption analysis can be called up for the measures from complaint management.



Calling up of referenced complaints

Function authorization: cm.*

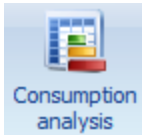
Calls up the complaint referenced for the selected measure.



Calling up of referenced inspection requirements

Function authorization: irp.*

Calls up the inspection requirement referenced for the selected measure.



Calling up of the consumption analysis

Function authorization: cona.*

Calls up the "consumption analysis" application

Editing functions

New measures can be created, provided that relevant skills are available. Consequently, users must have profound knowledge, as they are not guided through the process of data collection/modification and validation checking is not performed.

Relevant data need to be entered manually in the key fields 1 to 5 as well as in the context, type and area fields to enter new measures. The field descriptions provide further details about these field functions.

Please note:

New measures for complaints should be created within Complaint Management, as the creation within the Measure Tracking function requires the internal HYDRA complaint number to be entered.